.NET Skill Challenge

Add to your personal **GitHub** repository a new project for evaluation and send the URL to the recruiters for them to share it with the technical staff. After the technical interview you can delete it.

Develop a .NET application to be integrated to the proposed Angular application.

Car Rental

The car rental system at the airport will allow users to manage rentals efficiently and personally. A customer can check the availability of cars based on the date and desired features.
Register a rental by selecting a start date, specifying the vehicle characteristics they want (such as type and model), and defining the rental period by setting a return date.
In this way, the system fulfills the processes of booking, consultation, modification, and cancellation.
To complete a Rental, the following information must be known about the Customer: ID, full name, address, rental start and end days, and car type. The system will show options according to availability.
A car can only be assigned to one customer at a time.

```
class RentalSystem {
    +registerCustomer()
    +checkAvailability()
    +registerRental()
    +cancelRental()
}

class Customer {
    +ID: String
    +fullName: String
    +address: String
    +email: String
}

class Rental {
    +customer: Customer
    +startDate: Date
    +endDate: Date
    +car: Car
}
```



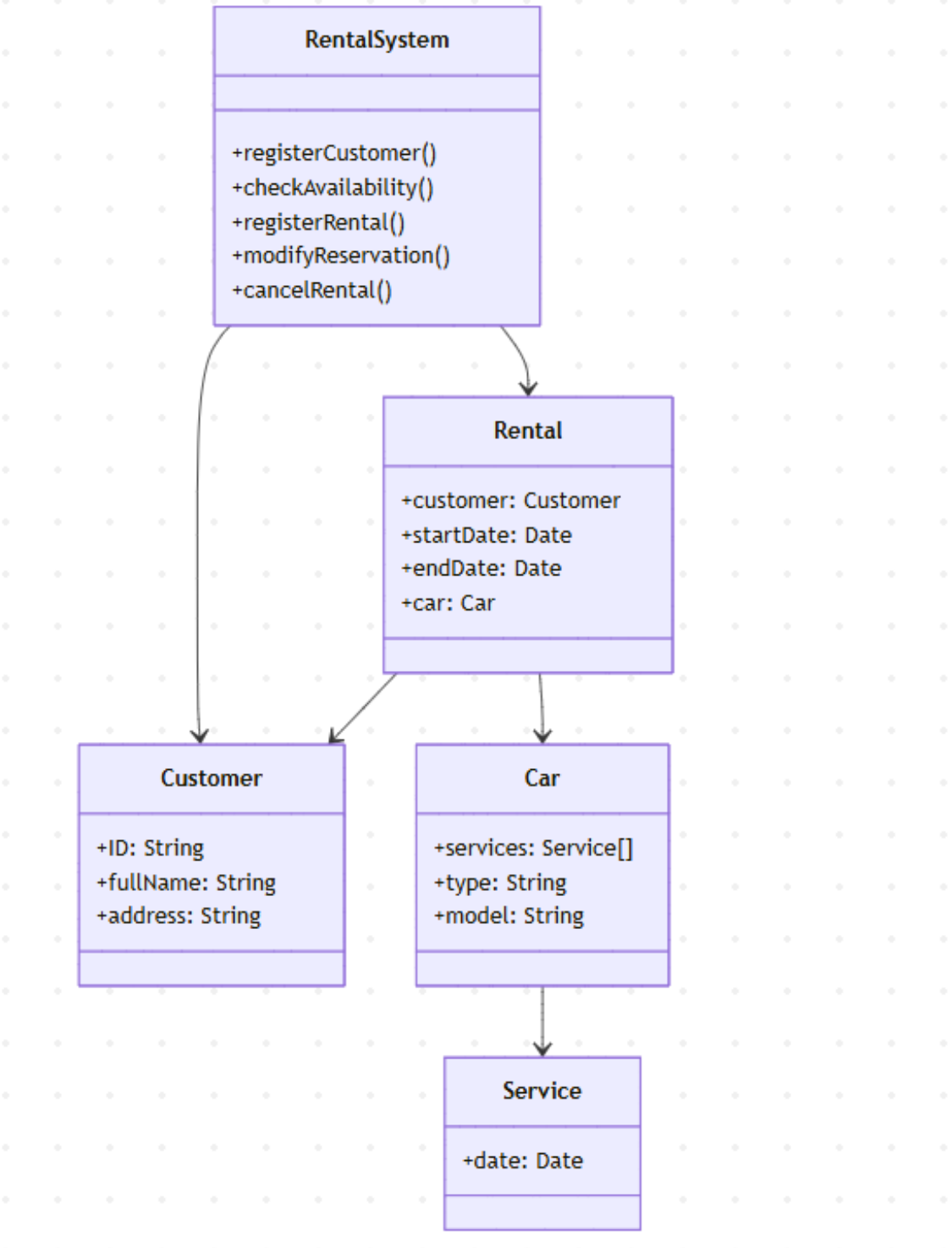
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```
class Car {
    +services: Service[]
    +type: String
    +model: String
}

class Service {
    +date: Date
}

RentalSystem --> Customer
RentalSystem --> Rental
Rental --> Customer
Rental --> Car
Car --> Service
```

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Here is a little and brief explanation about the following requirements for the **challenge**: In our structure, each layer of seniority includes specific responsibilities that build upon those of the previous tier. For example, a junior team member will have certain tasks, while a semi-senior team member will be expected to perform both the junior tasks and their own semi-senior responsibilities.

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Technical requirements:

- Implement a global exception handler
- Implement software architecture. For example.: DDD + CQRS
- Implement at least one design pattern (Excluding DI)
- Implement dependency injection (DI)
- Push descriptive atomic commits (See conventional commits)
- Push at least 10 commits
- The application must run correctly

Technical nice-to-haves:

- Implement in-memory cache & cache invalidation
- Dockerize the application
- Add authentication
- Add authorization
- Implement soft delete

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Agentic tools:

The use of AI-assisted development tools is allowed and encouraged as part of this challenge. You may use AI as a collaborative tool during the development process, for example, explore alternatives, generate or review code, create tests, troubleshooting issues, or improve documentation.

If you use AI, include a brief section in the README describing your AI-assisted development process. The purpose is not to evaluate how much AI-generated code was used, but to understand your workflow, technical judgment, and how you validated the proposed solutions.

Please mention, when applicable:

- Which AI tools, models, or coding assistants you used.
- What you used them for during the challenge.
- Your general workflow when interacting with these tools.
- Any custom instructions, rules, skills, agents, or MCP servers you configured.
- Examples of suggestions you accepted, modified, or rejected.
- How you reviewed, tested, and validated AI-assisted code.

You do not need to include complete conversation histories or disclose private information, credentials, API keys, or proprietary configurations.

The candidate remains responsible for the final implementation and should be able to explain and justify all submitted code and technical decisions during the interview.

If you did not use AI tools, simply mention it in the README and briefly describe your development process.